

Distributed Systems

Lecture1: Introduction and What Makes Systems Distributed?

Mohammad Reza Shakournia

shakournia@ut.ac.ir

School of Electrical and Computer Engineering,
University of Tehran,
Tehran, Iran

Why This Course?

- Distributed systems are everywhere.
- Bigger Problems like weather forecast, Economic modeling, Scientific problems, etc. can not be solved by a single machine.
- Some application are inherently distributed, sensor networks, etc.
- Resource sharing: A lot of Data to store in one place
- Solution: Distributed computing
- We need good knowledge on DS inside the country.

Learning Outcomes

① Explain core DS concepts: processes, communication, time, consistency

② Identify and handle faults, partial failures, and concurrency

③ Compare architectures: client-server, P2P, microservices, cloud/edge

④ Design systems with goals: scalability, dependability, security

⑤ Evaluate trade-offs using real-world case thinking

Course Roadmap

- Introduction and Design Goal of Distributed Systems
- Architectures
- Processes
- Communication
- Coordination
- Naming
- Consistency and Replication
- Fault Tolerance
- Security (Maybe)

References

① Distributed Systems, Tanenbaum & van Steen, 4th Edition ,2023.

② Coulouris, et al., “Distributed Systems: Concepts and Design”, 5th Edition.

Grading

Type	Ratio	Count	Score
Homeworks	25%	5	5
Paper Review	5%	4	1
Quizzes	10%	4	2
Midterm Exam	30%	1	6
Final Exam	30%	1	6
Bonus Project	15-20%	1	3-4
Total	100% + 15-20%	16	23-24

Midterm: 1405/02/07

Rules & Expectations

① No mobile phones or laptops are allowed during class sessions.

② Please do not copy homework/assignments from others.

③ If you use AI tools for coding:

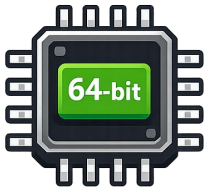
- ❑ Using AI tools (e.g., ChatGPT, Copilot) is allowed only as assistance.
- ❑ You must disclose AI use (what tool + where it was used, and include key prompts if asked).
- ❑ You are responsible for understanding and verifying the solution.
- ❑ During submission, you may be asked to explain your code/design; inability to explain may lead to loss of credit.

④ For collaboration in a group: Equal participation is required from all team members(Individual contribution may be checked).

The course is offered for the first time.
There will be try and errors. We welcome
any suggestions and comments.

Evolution of Computer Systems

- From 1945 to ~1985, computers were large, expensive, and mostly isolated.
- Without practical connectivity, early computers operated independently.
- From the mid-1980s, two key advances changed this situation:



Powerful Microprocessors rapidly evolving from 8-bit to 16/32/64-bit and multicore CPUs, delivering mainframe-level power at a tiny fraction of the cost.



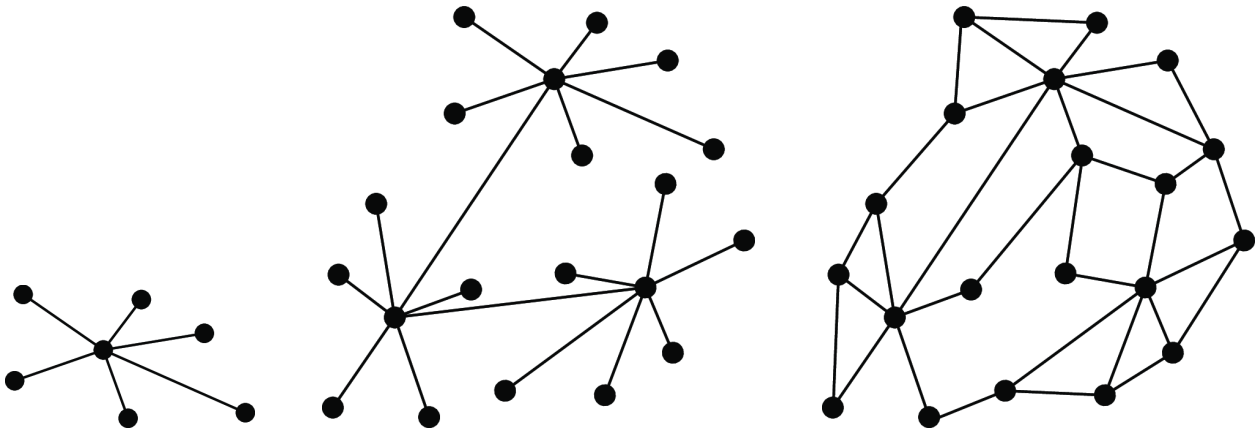
High-speed Networks LANs connect thousands of machines with microsecond transfers and multi-Gbps rates, while WANs link millions worldwide at widely varying speeds.

The World is Networked

- Computing has miniaturized into powerful connected devices, from smartphones to credit-card-sized boards (e.g., Raspberry Pi and Arduino)
- As society digitalizes, embedded networked computers now control critical infrastructure and even modern cars.
- Networked systems range from few to millions of devices and are highly dynamic.
- Since most systems are networked and Internet-accessible, understanding them is complex, so this lecture builds the big picture step by step.

Centralized vs Decentralized vs Distributed

- It is often claimed that centralized systems do not scale well.
- It is often claimed that distributed systems are more failure-robust,
- BUT THESE CLAIMS ARE NOT GENERALLY TRUE.



Centralized Decentralized Distributed

Centralized vs Decentralized vs Distributed

- A networked system can be seen from two main perspectives.
  - The **integrative view** connects existing systems so their services/resources can be shared in new ways (e.g., enterprise integration or scientific grid computing).
  - The **expansive view** extends a system with more computers for better performance and dependability (e.g., adding CDN edge servers to serve users faster).
- In both views, services run as distributed processes and resources across multiple computers, **leading to two types of networked systems**.
  - Decentralized System
  - Distributed System

Centralized vs Decentralized vs Distributed

Definition 1.1. Decentralized System

A decentralized system is a networked computer system in which processes and resources are **necessarily** spread across multiple computers.

Definition 1.2. Distributed System

A distributed system is a networked computer system in which processes and resources are **sufficiently** spread across multiple computers.

- Federated learning and Blockchain are examples of decentralized systems.
- Google email service, CDN, and NAS are examples of distributed systems.

The Distinction Matters!

- Centralized solutions are generally much simpler.
- Decentralization, that is, the act of spreading the implementation of a service across multiple computers.
- Distributed and decentralized solutions are inherently difficult:
  - Lots of Dependencies,
  - Partial Failures,
  - Highly Dynamic,
  - Vulnerable to Security Attacks.

False Assumptions!

- Peter Deutsch summarized common beginner mistakes as a set of false assumptions people make when first building distributed applications:
  - The network is reliable
  - The network is secure
  - The network is homogeneous
  - The topology does not change
  - Latency is zero
  - Bandwidth is infinite
  - Transport cost is zero
  - There is one administrator

Perspectives of Distributed Systems

- In this course, we want to take a look at distributed systems from different perspectives.
  - Architectural Perspective,
  - Process Perspective,
  - Communication Perspective,
  - Coordination Perspective,
  - Naming Perspective,
  - Consistency And Replication Perspective,
  - Fault Tolerance Perspective,
  - Security Perspective.

Wrap-Up + What's Next

In the next lecture, we will discuss key design goals of distributed systems:

- Resource sharing,
- Transparency,
- Openness,
- Dependability,
- Security,
- Scalability.